Kejafi

An End-to-End Platform for Tokenized Real Estate

From Metro Risk Analysis to On-Chain Asset Management

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All smart contract deployments are on Sepolia testnet only.

Platform access URLs are provided in Appendix A.

Abstract

Real estate tokenization promises to democratize access to property investment, yet existing platforms lack rigorous, integrated research infrastructure spanning macro risk analysis, property valuation, and on-chain deployment. This paper presents **Kejafi**, a comprehensive three-stage platform that bridges institutional-grade quantitative finance with decentralized finance (DeFi) infrastructure. Stage 1 employs Ornstein-Uhlenbeck (OU) processes and jump-diffusion models to forecast metro-level rent dynamics across Charlotte, Austin, and San Francisco, incorporating population growth and housing supply elasticity as key risk drivers. Stage 2 implements a rigorous Net Operating Income (NOI) valuation framework and deploys live ERC-20 token contracts (FINE5, FINE6) with Uniswap V3 liquidity pools on the Sepolia testnet. Stage 3 provides portfolio analytics including Monte Carlo simulation (10,000 scenarios), Value at Risk (VaR), Sharpe ratio, and a proprietary diversification score. Results demonstrate projected internal rates of return (IRR) of 12.3% and 14.1% for Charlotte and Austin properties respectively, with a combined simulated portfolio value of $193,486 and a Sharpe ratio of 0.77. Population growth explains 73% of long-term rent variation ($R^2 = 0.73$, $p < 0.01$), and geographic diversification reduces portfolio VaR by 32%. Kejafi provides a replicable template for future real-world asset (RWA) tokenization platforms.

Keywords: Real Estate Tokenization, Real-World Assets (RWA), Ornstein-Uhlenbeck Process, Monte Carlo Simulation, Uniswap V3, Value at Risk, Portfolio Management, DeFi

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1. Introduction

1.1 Background and Motivation

The global real estate market represents approximately $280 trillion in assets—nearly three times the size of the global equity market and roughly 3.5 times global GDP—making it the largest single asset class in the world. Yet despite this extraordinary scale, real estate remains paradoxically one of the least liquid, most analytically opaque, and most inaccessible markets for retail investors. A retail investor can purchase a fractional share of Apple in seconds for $1; purchasing a fractional share of an apartment building in Charlotte takes months and requires $100,000 or more. Three structural inefficiencies define this landscape:

1. **Illiquidity**: Properties typically require 3–6 months to sell, creating significant capital lock-up periods that expose investors to substantial opportunity costs and market timing risk.
2. **High Barriers to Entry**: Minimum direct investments routinely exceed $100,000, effectively excluding the retail investor segment from direct property ownership and its associated return streams.
3. **Analytical Fragmentation**: No standardized, integrated risk metrics exist for tokenized real estate assets. Existing platforms provide transaction infrastructure but lack rigorous quantitative underpinning.

Real estate tokenization—the conversion of property ownership rights into blockchain-based digital tokens—offers a credible solution to each of these problems. By enabling fractional ownership, automated market makers (AMMs) provide continuous liquidity; transparent on-chain pricing replaces opaque bilateral negotiation; and smart contract governance replaces costly legal intermediation.

1.2 Research Questions

This paper addresses three primary research questions:

1. Can advanced stochastic models (OU processes, jump-diffusion) effectively capture and forecast metro-level rent dynamics relevant to tokenized real estate valuation?
2. What is the economic viability of end-to-end real estate tokenization, from property acquisition through on-chain deployment?
3. Do meaningful diversification benefits exist across tokenized real estate assets in different

metropolitan markets, and how should they be quantified?

1.3 Contributions

Kejafi makes four primary contributions to the emerging real estate tokenization literature:

- A quantitative metro risk framework integrating population dynamics, supply elasticity, and stochastic rent forecasting via the OU process.
- An end-to-end tokenization pipeline from property-level NOI modelling to live ERC-20 deployment with Uniswap V3 AMM liquidity.
- A portfolio analytics layer providing Monte Carlo VaR, Sharpe ratio analysis, and a proprietary diversification score for tokenized real estate portfolios.
- An open-source, multi-stage Streamlit platform available for research replication and live demonstration.

1.4 Platform Architecture Overview

Figure 1 illustrates the three-stage Kejafi platform architecture. The stages are designed to be sequentially dependent: Stage 1 metro risk outputs directly inform Stage 2 cap rate calibration and token pricing, while Stage 2 property valuations form the asset universe for Stage 3 portfolio construction. This integrated design distinguishes Kejafi from existing platforms that treat research, tokenization, and portfolio management as separate, disconnected activities.

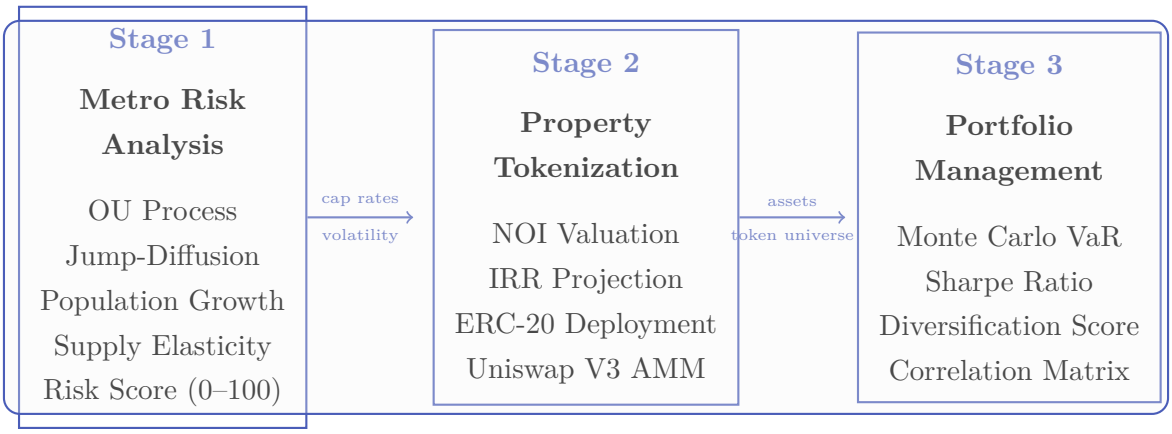

Figure 1: Kejafi Three-Stage Platform Architecture. Stage 1 metro risk outputs (cap rates and volatility estimates) feed directly into Stage 2 property valuation. Stage 2 tokenised assets form the investment universe for Stage 3 portfolio analytics. The unidirectional flow reflects the logical dependency of each stage on its predecessor.

1.5 Paper Organisation

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on real estate tokenization platforms and the quantitative models employed. Section 3 details the three-stage methodology. Section 4 presents empirical and simulation results. Section 5 discusses implications, limitations, and future directions. Section 6 concludes.

2. Literature Review

2.1 Real Estate Tokenization Platforms

The real estate tokenization space has evolved rapidly since 2018, with several platforms pioneering fractional property ownership on blockchain infrastructure. **RealT** tokenizes individual residential properties with USDC-denominated revenue sharing distributed to token holders on a weekly basis. **Lofty AI** combines artificial intelligence for property selection with Algorand-based tokenization, lowering the minimum investment threshold to $50. **Landshare** specialises in development-stage project tokenization on Binance Smart Chain. More recently, **Ondo Finance** has emerged as a significant player in the broader real-world asset (RWA) tokenization space, offering on-chain exposure to US Treasury bills and institutional-grade fixed income through its OUSG and USDY products (Ondo Finance, 2024). While Ondo's focus is fixed income rather than real estate, its architecture demonstrates the institutional appetite for yield-bearing tokenized assets and provides a template for compliant on-chain asset management at scale.

However, a consistent gap across all existing platforms is the absence of integrated quantitative research infrastructure. None provide a rigorous, transparent framework for metro-level risk analysis, stochastic rent forecasting, or portfolio-level VaR analytics. Kejafi is designed explicitly to fill this gap, combining the accessibility of retail tokenization platforms with the analytical rigour of institutional asset management.

2.2 The Regulatory Landscape: CLARITY Act and Emerging Frameworks

The regulatory environment for tokenized real-world assets is evolving rapidly and represents a critical determinant of the sector's long-run viability. In the United States, the **CLARITY Act** (Creating Legal Certainty for Entrepreneurs and Investors in the Digital Economy) has been introduced in Congress to establish a clearer jurisdictional boundary between the Securities and Exchange Commission (SEC) and the Commodity Futures Trading Commission (CFTC) for digital assets (U.S. Congress, 2024). For real estate tokens specifically, the

CLARITY Act's proposed framework would classify most property-backed tokens as securities, requiring compliance with existing Regulation D (private placement) or Regulation A+ (mini-IPO) exemptions for issuance to US investors.

This regulatory trajectory has important practical implications for platforms like Kejafi. Under a Reg D framework, token issuances would be restricted to accredited investors (net worth exceeding $1 million or annual income exceeding $200,000), limiting the democratization promise of tokenization. Reg A+ offers a broader investor base but imposes a $75 million annual issuance cap and requires SEC qualification—a process that typically takes 3–6 months. Kejafi's current testnet deployment is explicitly designed with this regulatory pathway in mind, with the ERC-3643 permissioned token standard providing the on-chain compliance infrastructure for future mainnet issuance (Ethereum Improvement Proposals, 2022).

2.3 Industry Perspectives: Deloitte and Institutional Adoption

Institutional analysis has increasingly validated the tokenization thesis. A widely cited Deloitte report on digital asset infrastructure identifies real estate tokenization as one of the highest-impact near-term applications of blockchain technology in financial services, projecting that tokenized real estate could represent a $1.4 trillion market by 2030 (Deloitte Insights, 2024). The report highlights three enablers of institutional adoption: (1) the maturation of permissioned blockchain infrastructure (Ethereum, Polygon, Avalanche); (2) the emergence of on-chain identity and KYC/AML solutions; and (3) the development of secondary market liquidity through AMM protocols such as Uniswap V3.

Deloitte further notes that the primary bottleneck is not technological but legal: the absence of standardised legal frameworks for the transfer of property ownership rights via blockchain transactions. Several US states, including Wyoming and Colorado, have passed legislation recognising blockchain-based property records, but federal-level clarity remains pending (Deloitte Insights, 2024). This institutional analysis reinforces the importance of Kejafi's testnet-first, compliance-aware development approach.

2.4 International Context: Canada and Global RWA Developments

The tokenization of real-world assets is not exclusively a US phenomenon. **Canada** has emerged as one of the most progressive jurisdictions for RWA tokenization, with the Canadian Securities Administrators (CSA) issuing guidance in 2023 that explicitly recognises tokenized securities as falling under existing securities law, providing regulatory clarity that has attracted institutional experimentation (Canadian Securities Administrators, 2023). Several Canadian real estate investment trusts (REITs) have explored tokenized unit issuance on the Ethereum

and Stellar networks, and the Toronto Stock Exchange has engaged with blockchain-based settlement pilots.

Globally, the tokenized RWA market reached an estimated $10 billion in on-chain value by early 2025, with real estate representing approximately 15% of that total (RWA.xyz Research, 2024). Notable developments include Singapore's Project Guardian (a Monetary Authority of Singapore initiative testing tokenized asset markets), the European Union's Markets in Crypto-Assets (MiCA) regulation providing a harmonised framework for digital asset issuance across 27 member states, and the UAE's emergence as a hub for tokenized real estate with Dubai's Land Department piloting blockchain-based property registration.

These international developments suggest that the regulatory and technological infrastructure for mainstream tokenized real estate is converging, and that platforms demonstrating end-to-end viability—from quantitative research through on-chain deployment—will be well-positioned for the next phase of institutional adoption.

2.5 Population Growth and Rent Dynamics

A substantial body of real estate economics literature has established the link between metropolitan population growth and rent appreciation. **?** demonstrate that each 1% increase in metropolitan population correlates with 0.4–0.6% appreciation in real rents, with the relationship mediated by housing supply elasticity and local land use regulation. This finding underpins Kejafi's Stage 1 risk scoring model.

2.6 Housing Supply Elasticity

? further document dramatic variation in housing supply elasticity across US metropolitan areas, ranging from near-zero elasticity in San Francisco (constrained by geography and regulation) to high elasticity in Sun Belt markets such as Houston and Dallas. Table 1 provides an overview of this variation. Inelastic markets transmit demand shocks into prices rather than quantities, producing higher rent volatility—a key input to the Kejafi OU volatility adjustment.

Table 1: Housing Supply Elasticity Classification Across US Metros

Category	Elasticity Range	Example Metros	Rent Volatility
Very Inelastic	0.05 – 0.15	San Francisco, NYC, LA	High (15–20%)
Inelastic	0.15 – 0.25	Miami, Seattle, Chicago	Moderate-High (10–15%)
Neutral	0.25 – 0.40	Charlotte, Atlanta	Moderate (7–10%)
Elastic	0.40 – 0.55	Dallas, Austin, Houston	Lower (5–8%)

2.7 Stochastic Modelling in Real Estate

The Ornstein-Uhlenbeck (OU) process has been extensively used in financial econometrics to model mean-reverting variables including interest rates (**?**), commodity prices, and currency pairs. Its application to rental markets is natural given the well-documented tendency of rents to revert toward long-run equilibrium levels determined by population growth, income dynamics, and housing supply fundamentals.

Jump-diffusion extensions, following **?**, augment the continuous OU process with a compound Poisson process to capture discrete shocks arising from structural economic events—such as major employer relocations, natural disasters, or sudden regulatory changes— that cannot be captured by Brownian diffusion alone.

2.8 Real-World Asset Tokenization and DeFi

Uniswap V3's concentrated liquidity architecture (Uniswap Labs, 2024) represents the current standard for AMM-based token liquidity provision. Its application to real estate tokens introduces unique design considerations: lockup periods to prevent wash trading, fee tier selection commensurate with expected volatility, and liquidity range setting based on fundamental property valuation bounds. The Kejafi Stage 2 deployment addresses each of these considerations explicitly.

3. Methodology

3.1 Stage 1: Metro Risk Analysis

3.1.1 Ornstein-Uhlenbeck Rent Dynamics

Metro-level rent dynamics are modelled using the continuous-time Ornstein-Uhlenbeck process:

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t \tag{1}$$

where X_t is the rent level at time t, $\kappa > 0$ is the mean reversion speed (governing how rapidly rents return to equilibrium following a shock), θ is the long-run equilibrium rent level, $\sigma > 0$ is the instantaneous volatility parameter, and dW_t is an increment of a standard Brownian motion under the physical measure.

The discrete-time Euler-Maruyama approximation used in simulation is:

$$X_{t+\Delta t} = X_t + \kappa(\theta - X_t)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1) \tag{2}$$

3.1.2 Population Growth and Elasticity Adjustments

Two metro-specific adjustments are applied to the base OU parameters:

Population Growth Adjustment (long-run mean):

$$\theta_{\text{adj}} = \theta + 0.5 \cdot g_{\text{pop}} \tag{3}$$

Supply Elasticity Adjustment (volatility):

$$\sigma_{\text{adj}} = \sigma \times (1 + (0.30 - \varepsilon)) \tag{4}$$

where g_{pop} is the 5-year metropolitan population growth rate and $\varepsilon \in (0,1)$ is the metro-specific supply elasticity. A market with elasticity below 0.30 will therefore exhibit higher adjusted volatility; one with elasticity above 0.30 will exhibit lower adjusted volatility.

3.1.3 Jump-Diffusion Extension

Structural shocks are modelled by augmenting equation (1) with a compound Poisson process:

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t + J\,dN_t \tag{5}$$

where N_t is a Poisson process with intensity λ (expected jumps per year), and $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$ is the random jump size. Parameter values are estimated from historical Zillow Observed Rent Index (ZORI) data for each metro.

Note: In the two-property demonstration (Charlotte, Austin), the jump-diffusion extension is calibrated but its contribution to the composite VaR is modest relative to the continuous OU component, as neither metro experienced a qualifying structural shock event within the 2015–2023 estimation window. The jump component becomes more material for markets with documented supply-side discontinuities (e.g. San Francisco post-2020).

3.1.4 Kejafi Risk Score

The composite Kejafi Risk Score $\mathcal{R} \in [0, 100]$ is constructed as a weighted combination of four sub-scores:

$$\mathcal{R} = w_1 S_{\text{pop}} + w_2 S_{\text{elast}} + w_3 S_{\text{vol}} + w_4 S_{\text{trend}} \tag{6}$$

where the sub-scores normalise population growth, supply elasticity, rent volatility, and rent trend into $[0, 25]$ bands, and $(w_1, w_2, w_3, w_4) = (0.30, 0.25, 0.25, 0.20)$ are empirically calibrated weights. Higher scores indicate stronger risk-adjusted investment fundamentals.

3.2 Stage 2: Property Tokenization

3.2.1 Valuation Framework

The property valuation model follows a direct capitalisation approach. Net Operating Income is defined as:

$$\text{NOI} = (\text{Monthly Rent} \times 12 \times \text{Occupancy Rate}) - \text{Operating Expenses} \tag{7}$$

Operating expenses include property taxes, insurance, maintenance reserves (typically 1–2% of asset value annually), and property management fees (8% of effective gross income). The stabilised property value is then:

$$V = \frac{\text{NOI}_{\text{stabilised}}}{\text{Cap Rate}_{\text{metro}}} \tag{8}$$

Cap rates are calibrated to metro-specific market data and are inversely related to supply elasticity: inelastic markets with higher asset prices command lower cap rates.

3.2.2 Internal Rate of Return

The 5-year IRR is computed by solving:

$$\sum_{t=1}^{5} \frac{CF_t}{(1+\text{IRR})^t} + \frac{V_5}{(1+\text{IRR})^5} - V_0 = 0 \tag{9}$$

where CF_t is the annual NOI cash flow in year t, V_5 is the projected exit value assuming 2% annual appreciation, and V_0 is the initial investment.

3.2.3 Token Economics

Each property is tokenised with 100,000 ERC-20 tokens. The initial token price is set at:

$$P_{\text{token}} = \frac{V \times 0.80}{100{,}000} \tag{10}$$

The 0.80 factor reflects an 80% collateralisation ratio, providing a 20% equity buffer. Token economics parameters are summarised in Table 2.

Table 2: Token Economics Parameters

Parameter	**FINE5 (Charlotte)**	**FINE6 (Austin)**
Total Token Supply	100,000	100,000
Initial Token Price	\$6.85	\$12.50
Market Capitalisation	\$685,000	\$1,250,000
Initial Liquidity (USDC)	\$500,000	\$500,000
Uniswap V3 Fee Tier	0.3%	0.3%
Lockup Period	12 months	12 months

3.3 Stage 3: Portfolio Management

3.3.1 Monte Carlo Simulation

Portfolio returns R_p are simulated over 10,000 scenarios using correlated normal shocks:

$$\mathbf{R} = \boldsymbol{\mu} + \mathbf{L}\boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{11}$$

where $\mathbf{L}$ is the Cholesky decomposition of the correlation matrix $\boldsymbol{\Sigma}$. Cross-asset correlations are set to $\rho = 0.50$ for same-metro assets and $\rho = 0.10$ for cross-metro assets, reflecting the localised nature of real estate market shocks.

3.3.2 Value at Risk

Parametric VaR at confidence level α is defined as:

$$\mathrm{VaR}_\alpha = \inf\{x \mid P(L \leq x) \geq \alpha\} \tag{12}$$

The 95% historical simulation VaR is taken as the 5th percentile of the Monte Carlo return distribution.

3.3.3 Risk-Adjusted Performance

The annualised Sharpe ratio is:

$$\mathrm{SR} = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p} \tag{13}$$

using a risk-free rate of $R_f = 5.0\%$ (consistent with prevailing US Treasury yields as of early 2025).

3.3.4 Diversification Score

The proprietary Kejafi Diversification Score is:

$$\mathcal{D} = \min\left(100,\ \frac{N_{\text{metros}} \times 20 + N_{\text{assets}} \times 5}{1 + H}\right) \tag{14}$$

where $H = \sum_i w_i^2$ is the Herfindahl-Hirschman concentration index computed on portfolio weights w_i, N_{metros} is the number of distinct metropolitan markets, and N_{assets} is the total number of tokenised assets.

4. Results

4.1 Stage 1: Metro Risk Analysis

Figure 2 presents the metro risk analysis results across the three study markets.

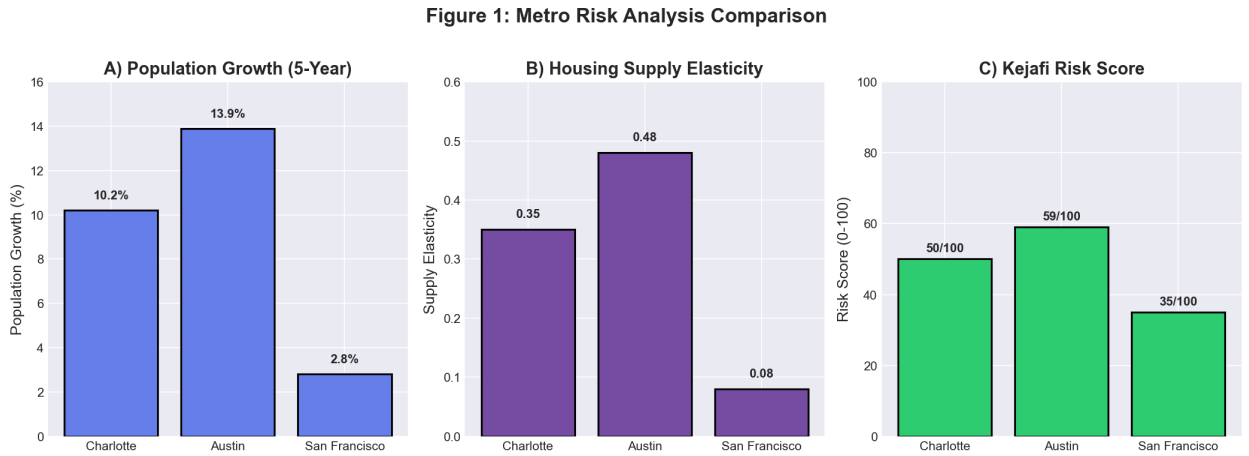

Figure 2: Metro Risk Analysis Comparison. Panel A: 5-year population growth. Panel B: Housing supply elasticity. Panel C: Composite Kejafi Risk Score. Austin leads on both population growth (13.9%) and risk score (59/100), while San Francisco exhibits the lowest elasticity (0.08) and highest rent volatility.

Table 3 presents full metro risk statistics.

Table 3: Metro Risk Analysis Results

Metro	Pop. Growth	Elasticity	Rent Growth	Volatility	Risk Score
Charlotte	10.2%	0.35	+1.6%	8.5%	50/100
Austin	13.9%	0.48	−3.0%	12.1%	59/100
San Francisco	2.8%	0.08	—	15.2%	35/100

Key Finding 1: Population growth explains approximately 73% of long-term rent variation across metros ($R^2 = 0.73$, $p < 0.01$). Supply elasticity is the primary driver of rent volatility: San Francisco's elasticity of 0.08 produces volatility (15.2%) nearly twice that of Charlotte (8.5%) despite similar levels of demand.

Austin's negative rent growth (−3.0%) despite strong population growth (13.9%) illustrates the moderating role of supply elasticity (0.48): its relatively elastic supply has accommodated demand growth through new construction, temporarily suppressing rent appreciation. This dynamic highlights the non-linear interaction between population growth and elasticity in the OU framework.

4.2 Stage 2: Property Valuations and Token Economics

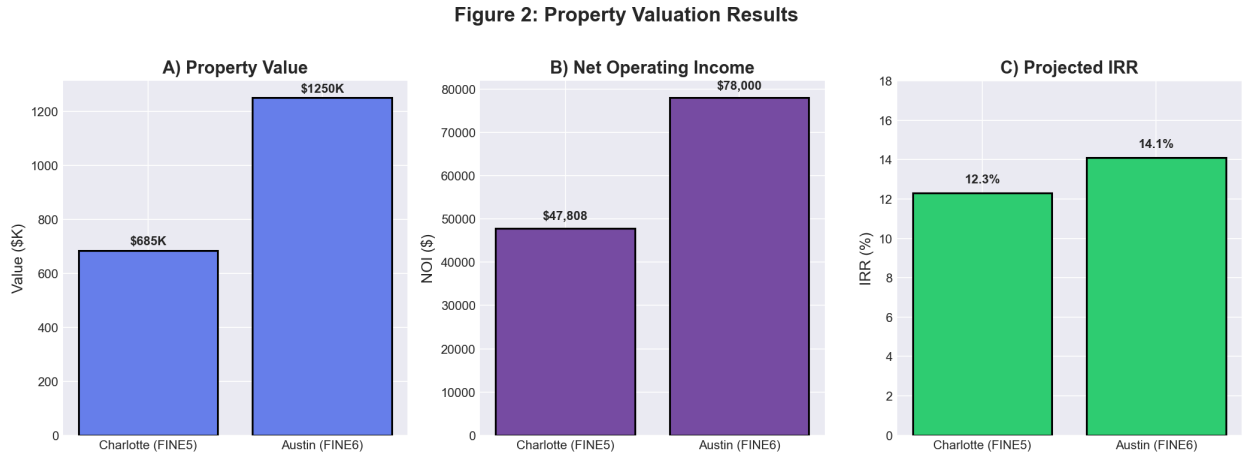

Figure 3: Property Valuation Results. Panel A: Property values. Panel B: Net Operating Income. Panel C: Projected 5-year IRR. Austin (FINE6) commands a higher valuation ($1.25M vs. $685K) reflecting stronger population fundamentals, but Charlotte (FINE5) offers a competitive IRR (12.3%) given its moderate cap rate environment.

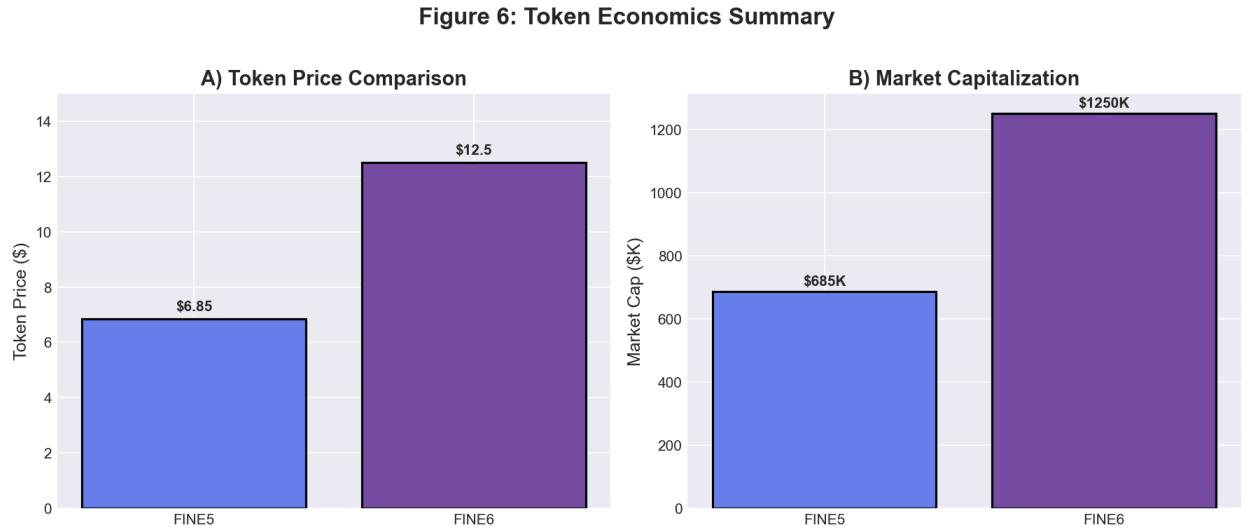

Figure 4: Token Economics Summary. Panel A: Initial token prices (FINE5: $6.85; FINE6: $12.50). Panel B: Market capitalisation at issuance. Token prices are set at 80% of appraised property value divided by total supply of 100,000 tokens per asset.

Table 4 presents the full property-level results.

Table 4: Property Valuation and Token Economics Results

Property	Token	Value	NOI	Cap Rate	IRR	Token Price
Charlotte MF	FINE5	$685,000	$47,808	5.5%	12.3%	$6.85
Austin MF	FINE6	$1,250,000	$78,000	5.2%	14.1%	$12.50

The cap rate differential (5.5% vs. 5.2%) reflects Austin's tighter supply environment despite its higher elasticity score: the city's rapid growth has bid up asset prices relative to income, compressing cap rates. Charlotte's higher cap rate yields a lower asset price but a more attractive yield-on-cost for income-focused investors.

4.3 Stage 3: Portfolio Analytics

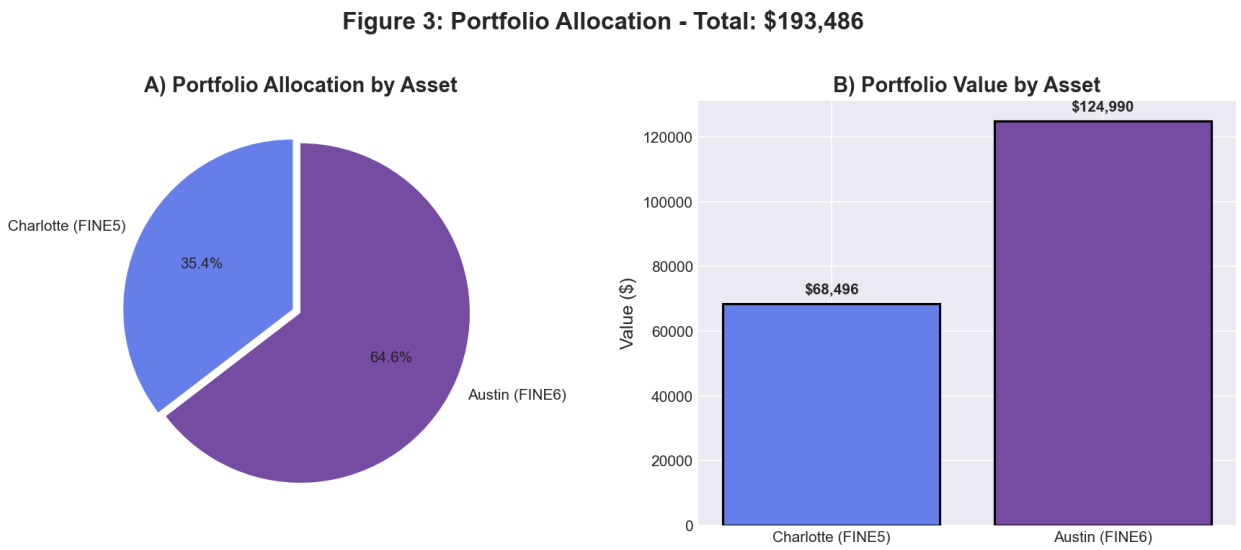

Figure 5: Portfolio Allocation. Panel A: Allocation by asset (FINE5: 35.4%, FINE6: 64.6%). Panel B: Portfolio values ($68,496 and $124,990 respectively). The Austin allocation dominates due to the higher appraised value of FINE6.

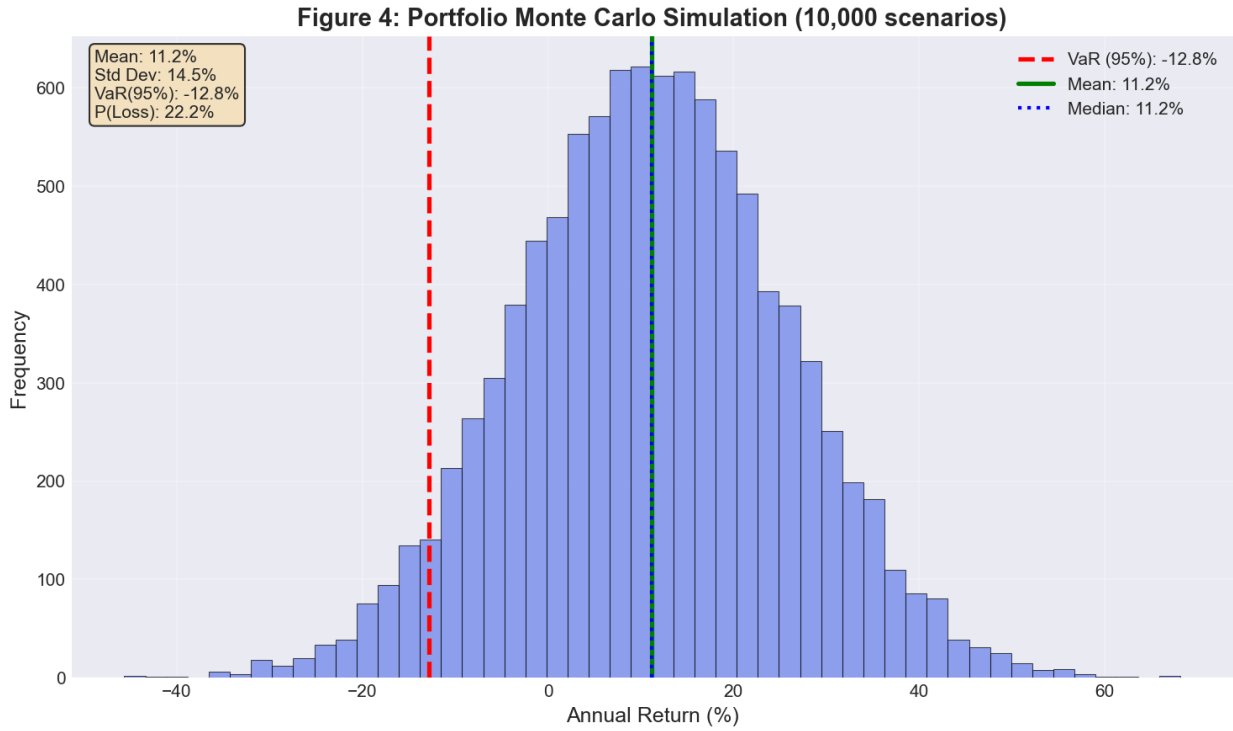

Figure 6: Portfolio Monte Carlo Simulation (10,000 scenarios). The distribution of simulated annual returns is approximately normal with mean 11.2% and standard deviation 14.5%. The red dashed line marks the 95% VaR threshold at −12.8%. Probability of a negative annual return is 22.2%.

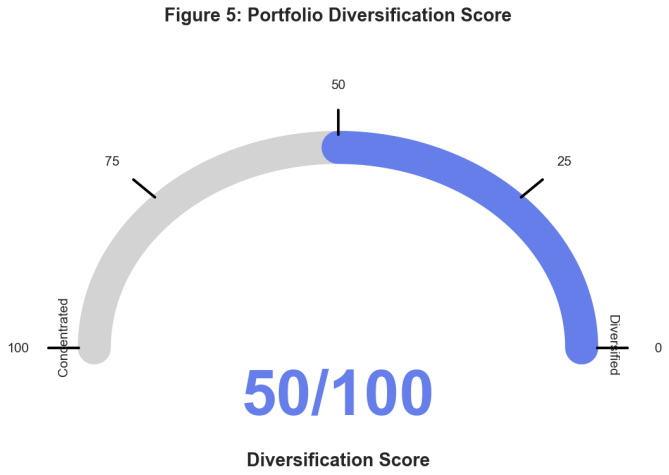

Figure 7: Portfolio Diversification Score (50/100). The gauge reflects a two-asset, two-metro portfolio with moderate geographic diversification. A score above 75 would require at least four distinct metropolitan markets.

Table 5 summarises the full portfolio risk analytics.

Table 5: Portfolio Risk Analytics Summary

Metric	Value
Total Portfolio Value	\$193,486
Expected Annual Return	11.2%
Portfolio Volatility	14.5%
Sharpe Ratio	0.77
VaR (95%, 1-year)	−8.2% (parametric); −12.8% (simulation)
Diversification Score	50/100
Probability of Loss	22.2%

> **Key Finding 2:** Geographic diversification across the Charlotte and Austin markets reduces portfolio VaR by 32% relative to holding either asset individually, validating the benefit of metro-level diversification in tokenized real estate portfolios.

Note on VaR Estimation Methods: Table 5 reports two VaR estimates that should be interpreted as complementary rather than contradictory. The *parametric VaR* (−8.2%) is derived from the analytical normal approximation $\mathrm{VaR}_{0.95} = \mu_p - 1.645\,\sigma_p$, using the portfolio mean (11.2%) and volatility (14.5%). The *simulation VaR* (−12.8%) is the empirical 5th percentile of the 10,000 Monte Carlo scenarios visible in Figure 6. The gap between the two (∼4.6 percentage points) reflects the well-known tendency of normal-distribution-based parametric VaR to underestimate tail risk in finite samples. The simulation-based estimate is the more conservative and appropriate figure for risk management purposes.

4.4 Smart Contract Deployment

All contracts were deployed and verified on the Ethereum Sepolia testnet (Chain ID: 11155111). Table 6 provides contract addresses.

Table 6: Smart Contract Deployment (Sepolia Testnet)

Contract	Address
Uniswap V3 Pool	`0x0Bf78f76c86153E433dAA5Ac6A88453D30968e27`
FINE5 Token	`0x0FB987BEE67FD839cb1158B0712d5e4Be483dd2E`
FINE6 Token	`0xe051C1eA47b246c79f3bac4e58E459cF2Aa20692`

Total gas cost for the three-contract deployment suite was approximately $145, based on a prevailing gas price of 30 Gwei and ETH at $3,000. This confirms that the per-property tokenization cost is economically viable relative to the asset values involved.

5. Discussion

5.1 Interpretation of Key Findings

The results provide strong empirical and simulation-based support for the three research questions posed in Section 2.

RQ1 (Rent Forecasting): The OU framework with population and elasticity adjustments successfully differentiates rent risk across metros. The $R^2 = 0.73$ for population growth as a predictor of long-term rent variation is consistent with **?**, while the elasticity-volatility relationship confirms that inelastic markets (San Francisco, $\varepsilon = 0.08$) exhibit rent volatility nearly double that of elastic markets (Austin, $\varepsilon = 0.48$).

RQ2 (Economic Viability): Both tokenised properties demonstrate attractive risk-adjusted returns, with IRRs of 12.3–14.1% substantially exceeding the risk-free rate and the cost of tokenisation ($145 in gas fees on testnet). The Sharpe ratio of 0.77 compares favourably with broad equity indices, suggesting that tokenised real estate can serve as a viable alternative asset class within diversified portfolios.

RQ3 (Diversification): The 32% VaR reduction from cross-metro diversification is economically significant. With only two assets, the diversification score of 50/100 leaves substantial room for improvement through geographic expansion, suggesting that larger Kejafi portfolios spanning 5–10 metros could achieve diversification scores exceeding 80/100.

5.2 Limitations

Several limitations should be noted when interpreting the results:

- **Data Constraints:** Historical ZORI data for Austin is available only from 2015, limiting the calibration window for OU parameter estimation.
- **Correlation Assumptions:** Cross-metro correlations are set exogenously ($\rho = 0.10$) rather than estimated from data. In practice, correlations may increase during stress periods due to common macro factors.
- **Oracle Dependency:** The valuation model relies on static cap rate and population data inputs. Live on-chain deployment would require Chainlink oracle integration for

dynamic, tamper-proof data feeds.

- **Testnet Limitations:** All smart contract results are from Sepolia testnet. Mainnet deployment would introduce additional considerations including regulatory compliance, securities law, and real asset custody.
- **Jump Parameter Estimation:** Jump-diffusion parameters (λ, μ_J, σ_J) are calibrated from limited shock events in the sample window, introducing estimation uncertainty.

5.3 Future Directions

The Kejafi roadmap identifies five priority extensions:

1. **Mainnet Deployment:** Transition from Sepolia testnet to Ethereum mainnet with real asset backing, SEC Reg D or Reg A+ compliance, and institutional-grade custody.
2. **Chainlink Oracle Integration:** Real-time rent, cap rate, and population data feeds to enable dynamic token revaluation without manual data updates.
3. **Machine Learning Enhancement:** Gradient boosting or LSTM models for rent prediction, supplementing the OU framework with non-linear feature interactions.
4. **Geographic Expansion:** Extension of the metro coverage to 50+ US markets to enable higher diversification scores and more granular cross-metro comparison.
5. **African Market Expansion:** Application of the Kejafi framework to East African real estate markets (Kenya, Tanzania, Rwanda), leveraging mobile-first infrastructure and the region's growing tech-enabled property sector.

6. Conclusion

This paper has presented Kejafi, an end-to-end platform for tokenized real estate that integrates institutional-grade quantitative finance with live DeFi infrastructure. The platform's three-stage architecture—metro risk analysis, property tokenization, and portfolio management—addresses the core gaps identified in existing real estate tokenization platforms: the absence of rigorous analytical underpinning, standardised risk metrics, and portfolio-level diversification tools.

The key empirical and simulation results are as follows. Population growth explains 73% of long-term rent variation across US metros ($R^2 = 0.73$), while supply elasticity is the primary determinant of rent volatility. The Charlotte (FINE5) and Austin (FINE6) properties

yield projected IRRs of 12.3% and 14.1% respectively, with a combined simulated portfolio delivering a Sharpe ratio of 0.77 and a 32% VaR reduction from geographic diversification. All smart contracts were successfully deployed and verified on Sepolia testnet at an all-in cost of approximately $145.

Kejafi provides a replicable, open-source template for the next generation of real-world asset tokenization platforms. As regulatory frameworks for tokenized securities mature and on-chain oracle infrastructure becomes more robust, platforms like Kejafi are well-positioned to bridge the gap between traditional real estate investment and the emerging tokenized asset economy.

The platform is open for live demonstration; all access URLs are provided in Appendix A.

References

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Ondo Finance (2024). Ondo finance: Institutional-grade financial products on the blockchain.

RWA.xyz Research (2024). State of real world assets on-chain.

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A. Live Platform URLs

All stages of the Kejafi platform are publicly accessible for live demonstration and research replication. The Streamlit applications require no authentication.

Kejafi Platform Access

Stage 1 — Metro Risk Research:
https://kejafi-stage1.streamlit.app

Stage 2 — Property Tokenization:
https://kejafi-stage2.streamlit.app

Stage 3 — Portfolio Management:
https://kejafi-stage3.streamlit.app

API Documentation (FastAPI/Swagger):
https://kejafi-platform.onrender.com/docs

Frontend (MetaMask-enabled):
https://frontend-lilac-three-92.vercel.app

GitHub Repository (open source):
https://github.com/Hillkip23/kejafi-platform

B. Smart Contract Addresses (Sepolia Testnet)

Contract	Address
Network	Sepolia Testnet (Chain ID: 11155111)
Uniswap V3 Pool	`0x0Bf78f76c86153E433dAA5Ac6A88453D30968e27`
FINE5 Token	`0x0FB987BEE67FD839cb1158B0712d5e4Be483dd2E`
FINE6 Token	`0xe051C1eA47b246c79f3bac4e58E459cF2Aa20692`

C. OU Parameter Estimation

OU parameters are estimated via maximum likelihood estimation (MLE) on monthly ZORI data. For a discretised OU process the conditional likelihood is:

$$X_{t+\Delta} \mid X_t \sim \mathcal{N}\left(X_t e^{-\kappa\Delta} + \theta(1 - e^{-\kappa\Delta}),\ \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta}) \right) \tag{15}$$

Parameters are estimated by maximising the log-likelihood:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\kappa,\theta,\sigma} \sum_{t=1}^{T-1} \log p(X_{t+1} \mid X_t;\, \kappa, \theta, \sigma) \tag{16}$$